Patch-Based Diffusion Models for Solving Inverse Problems in Computed Tomography Imaging

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Abstract

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1 Introduction

Computed Tomography (CT) imaging is widely used in both medicine and industry for investigating the internal structure of objects in a non-destructive manner. It works by directing X-rays through a sample from a number of different angles, and measuring their attenuation at each angle. In the medical domain, we wish to minimise the radiation dose to the patient, so measurements are conducted at as few angles (views) as possible and with the lowest possible X-ray energy. This leads to artifacts from both Gaussian and Poisson noise, as well as from the limited number of views restricting the attainable information, making it difficult to accurately reconstruct the internal structure of the sample. Although industrial applications are often able to use higher energy X-rays and use many more views, there are still many cases where radiation dosage can damage the sample, or where only a limited number of views are possible due to movement [].

In this work, we aim to reproduce work by Ho et. al. Ho et al. [2], which introduces a method for improving the quality of very low-dose CT reconstructions by using a patch-based diffusion (PaDIS) model as a prior. The reproduction of CT and machine learning research is often difficult due to the relatively limited number of public CT datasets, concerns regarding patient privacy, and poorly engineered or maintained codebases. For this reason, we developed our reproduction natively within the LION framework [**?**], which provides physically accurate CT projection operators, standardised datasets and metrics, and a well-engineered codebase for developing and testing machine learning methods for CT reconstruction. We have therefore oped to maximise the utility of this work and enable easy comparison to future work by using a default LION dataset and geometry for our reproduction.

As well as implementing the core PaDIS pipeline, we also re-implemented the patch-averaging [3] and patch-stitching [4] inspired variants discussed in the PaDIS paper. Furthermore, we also ran experiments with some of the methods already implemented in LION, choosing those which are similar to the comparison models used by the original PaDIS paper.

The full methodology of our reproduction is described in **??**, with the results presented in **??**. These are then discussed in **??**, along with comments on the reproducibility of the Hu et. al. paper, before conclusions are presented in **??**. A theoretical overview of CT reconstruction and the variance exploding noise conditional score model (NCSN) used by PaDIS now follows.

2 Background

Here, we present a summary of the theory and literature required to understand the experiments presented in this work. We focus on CT reconstruction, diffusion models and their use in solving inverse problems, and the PaDIS method. We then summarise the literature relevant to PaDIS, including both its motivators and derivatives. The formulation presented in 2.1 summarises the key points presented by Kak and Slaney [**?**], and utilises their notation.

2.1 Computed Tomography

In X-ray CT, an object is illuminated by X-rays from multiple angles, with the intensity of radiation transmitted through the object measured by detectors the opposite side of the object. Through this method, we aim to reconstruct a spatial distribution of the linear attenuation coefficient, written $f(x,y)$ in the two-dimensional case, and with units of inverse length. This gives the probability per unit path length that an X-ray photon is removed from the beam by absorption or scattering. Therefore, assuming X-rays are monoenergetic, the transmitted intensity along a ray L should obey the Beer-Lambert law [],

$$I = I_0 \exp\left(-\int_L f(x,y)\, ds\right), \tag{2.1}$$

where I_0 is the intensity of radiation incident on the object, I is the detected intensity, and ds denotes integration along the ray. Taking logarithms therefore then gives the linearised projection

$$P = -\log\left(\frac{I}{I_0}\right) = \int_L f(x,y)\, ds, \tag{2.2}$$

meaning log-normalised CT measurements are approximated by a line integral through the attenuation image. This approach assumes that beams are narrow, monoenergetic and non-diffracting, and neglects scattering, beam hardening and finite aperture effects, all of which contribute in clinical CT. However, it provides a tractable mathematical basis within which to consider reconstruction algorithms.

2.1.1 Detector Geometries

There are three common CT detector geometries; parallel-beam, fan-beam, and cone-beam. Parallel-beam assumes we have a linear array of X-ray sources, each focussed solely on a detector placed on the other side of the sample. These rays are therefore parallel, as illustrated in Figure **??**a. This is impractical to implement clinically, but allows for tractable mathematics for the Radon transform and the Fourier slice theorem.

Fan-beam CT is often used in practice, and utilises a point source from which a fan-blade of emissions are measured by an line of detectors. Writing the source position as $S(\beta)$ for a source angle β, and $d(\beta,\gamma)$ as the unit direction of a ray labelled by fan angle γ allows us to write a fan-beam projection as

$$R_\beta(\gamma) = \int_0^\infty f(S(\beta) + sd(\beta,\gamma))\,ds, \tag{2.3}$$

where we assume the sample has compact support such that $f(\mathbf{r}) = 0$ outside of the object.

Cone-beam CT extends this to three dimensions, again using a point source, but now a two-dimensional array of detectors with coordinates labelled u and v, such that the cone-beam projection has the form

$$R_\beta(u,v) = \int_0^\infty f(S(\beta) + sd(\beta,u,v))\,ds. \tag{2.4}$$

Note, however, that by restricting the cone-beam geometry to a single strip of detectors recovers the fan-beam geometry.

2.1.2 Line Integrals, Projections and Sinograms

For a fixed projection angle θ, we can define a line in the image plane by

$$x\cos\theta + y\sin\theta = t, \tag{2.5}$$

where t is the perpendicular distance of the line from the origin, as shown in Figure **??**. The integral over this line can then be denoted as

$$P_\theta(t) = \int_{L_{(\theta,t)}} f(x,y)\,ds, \tag{2.6}$$

or equivalently in Dirac delta form,

$$P_\theta(t) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\delta(x\cos\theta + y\sin\theta - t)\,dx\,dy. \tag{2.7}$$

Notably, the projection $P_\theta(t)$ is therefore also the Radon transform [**?**] of $f(x,y)$ at angle θ. We collect a line or grid of these projections simultaneously, each corresponding to the same θ alignment, but different values of t. In a hypothetical parallel-beam scan, all of these θ-aligned rays are parallel, whereas in a realistic fan-beam or cone-beam scan, the angle refers to the central projection angle, from which other rays diverge.

In practice, we sample these projection lines from N_θ different angular views, $\theta_1, \theta_2, \ldots, \theta_{N_\theta}$, and collect the projections within N_t detector bins corresponding to $t_1, t_2, \ldots t_{N_t}$, forming an array of measurements called a sinogram, denoted

$$Y_{j,i} = P_{\theta_i}(t_j), \quad i = 1,\ldots,N_\theta, \quad j = 1,\ldots,N_t. \tag{2.8}$$

Here, each column is a line of projections at directed angle θ_i, and each row corresponds to one detector bin. By vectorising the image of $f(x,y)$ as $\mathbf{x} \in \mathbb{R}^n$, and the sinogram as $\mathbf{y} \in \mathbb{R}^m$, the discretised CT forward model can be written as

$$\mathbf{y} = A\mathbf{x} + \eta, \tag{2.9}$$

where $A \in \mathbb{R}^{m\times n}$ is the discretised forward projection operator, η encodes noise and modelling errors, and $m = N_\theta N_t$ in 2D. We can then express this operator A explicitly by denoting an image pixel ℓ by Ω_ℓ, and using the pixel basis function

$$\varphi_\ell(x,y) = \begin{cases} 1, & (x,y) \in \Omega_\ell, \\ 0, & \text{otherwise}, \end{cases} \tag{2.10}$$

such that we can approximate the attenuation field as

$$f(x,y) \approx \sum_{\ell=1}^{n} x_\ell \varphi_\ell(x,y), \tag{2.11}$$

where x_ℓ is the value of pixel ℓ. Therefore, substituting into (2.5) and then comparing with (2.9) gives

$$P_{\theta_i}(t_j) \approx \sum_{\ell=1}^{n} x_\ell \int_{L_{(\theta_i,t_j)}} \varphi_\ell(x,y)\,ds \tag{2.12}$$

$$\implies A_{(j,i),\ell} = \int_{L_{(\theta_i,t_j)}} \varphi_\ell(x,y)\,ds, \tag{2.13}$$

so each row of A corresponds to a single measured ray, with entries encoding the path length of that ray within each image pixel. This matrix is generally extremely large, and impractical to store or manipulate.

2.1.3 The Fourier Slice Theorem

Using the perpendicular distance of a ray from the origin, (2.5) and the distance along that ray given by

$$s = -x\sin\theta + y\cos\theta \tag{2.14}$$

give the projection at angle θ as

$$P_\theta(t) = \int_{-\infty}^{\infty} f(t\cos\theta - s\sin\theta,\ t\sin\theta + s\cos\theta)\,ds. \tag{2.15}$$

Taking the Fourier transform of this with respect to t then gives a Fourier slice,

$$S_\theta(\omega) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t\cos\theta - s\sin\theta,\ t\sin\theta + s\cos\theta)e^{-2\pi i\omega t}\,ds\,dt, \tag{2.16}$$

so converting to the (x,y) coordinate system gives

$$S_\theta(\omega) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)e^{-2\pi i\omega(x\cos\theta + y\sin\theta)}\,dx\,dy = F(\omega\cos\theta, \omega\sin\theta). \tag{2.17}$$

Therefore, the Fourier transform of a projection gives the two-dimensional Fourier transform of the object along the radial line at the same angle, as illustrated in Figure **??**.

2.1.4 Dosage and Noise

The radiation dose imparted on a patient is directly proportional to the total energy deposited in the tissue. For this reason, it is also directly proportional to the energy of the photons used, the number of photons released at each view, and the number of views acquired []. Dosage can therefore be reduced by decreasing any of these properties. However, these each decrease imaging quality.

Measurement noise is intrinsically linked to the intensity of radiation used. If C_0 photons are incident on the sample along a ray with projection value P, then we expect to detect

$$\bar{C} = C_0 \exp(-P). \tag{2.18}$$

As this is a counting experiment, the noise due to counting errors can be approximated as Poisson such that

$$C \sim \text{Poisson}(\bar{C}). \tag{2.19}$$

The projection estimate,

$$\hat{P} = -\log\left(\frac{C}{C_0}\right), \tag{2.20}$$

so at high photon counts where $\text{Poisson}(\bar{C}) \approx \mathcal{N}(\bar{C}, \bar{C})$, Taylor expansion gives

$$\hat{P} \approx P + \eta, \quad \eta \sim \mathcal{N}\left(0, \frac{1}{\bar{C}}\right), \tag{2.21}$$

at low photon counts (intensities), however, we may have $C = 0$, in which case $\hat{P} = -\log 0$, which is undefined. Furthermore, relative fluctuations are large, preventing Taylor expansion, and in general, the noise after the logarithm becomes non-gaussian, signal-dependent, biased and possibly undefined [], meaning this domain is often better to model with the Poisson likelihood. Decreasing intensity therefore increases the difficulty in modelling, and is not discussed in detail in this work. However, interesting advances in using diffusion models trained with these sorts of priors can be found at [].

Decreasing the number of views also decreases the dosage, but in accordance with the Fourier slice theorem, this also limits the angular coverage of the Fourier space corresponding to

$f(x,y)$, with only a finite set of radial Fourier lines being directly sampled. The unsampled angular regions must therefore be inferred by a reconstruction algorithm or a prior, motivating the use of the learned image priors discussed in Section 2.2. If using a direct reconstruction approach, however, this lack of information can lead to streak artefacts and poorly defined edges where high-frequency Fourier information is not available.

2.1.5 Hounsfield Units

Medical CT images are often displayed in Hounsfield units (HU) with restricted ranges []. These normalise attenuation values with respect to water and air such that

$$\mathrm{HU} = 1000\frac{\mu - \mu_{\mathrm{water}}}{\mu_{\mathrm{water}} - \mu_{\mathrm{air}}} \approx 1000\left(\frac{\mu}{\mu_{\mathrm{water}}} - 1\right), \tag{2.22}$$

giving water a value around 0 HU and air around -1000 HU.

2.2 Diffusion Models

A diffusion model is a constrained version of a Variational Autoencoder [] where a fixed encoder is used to gradually corrupt data, and a learned decoder acts to reverse this process. Where $\mathbf{x}$ denotes a clean data sample and $\mathbf{z}_1,\ldots,\mathbf{z}_T$ denote noisy latent variables, the forward process is

$$\mathbf{z}_1 = \sqrt{1-\beta_1}\mathbf{x} + \sqrt{\beta_1}\varepsilon_1, \tag{2.23}$$

$$\mathbf{z}_t = \sqrt{1-\beta_t}\mathbf{z}_{t-1} + \sqrt{\beta_t}\varepsilon_t, \qquad t = 2,\ldots,T, \tag{2.24}$$

where $\varepsilon_t \sim \mathcal{N}(\mathbf{0},\mathbf{I})$ and β_t is the discrete noise schedule. This can equivalently be written as

$$q(\mathbf{z}_t \mid \mathbf{z}_{t-1}) = \mathrm{Norm}_{\mathbf{z}_t}\left[\sqrt{1-\beta_t}\mathbf{z}_{t-1}, \beta_t\mathbf{I}\right]. \tag{2.25}$$

By defining

$$\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s) \tag{2.26}$$

and recursively combining Gaussian noise terms, we obtain the diffusion kernel

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t}\mathbf{x} + \sqrt{1-\bar{\alpha}_t}\varepsilon \tag{2.27}$$

$$\implies q(\mathbf{z}_t \mid \mathbf{x}) = \text{Norm}_{\mathbf{z}_t}\left[\sqrt{\bar{\alpha}_t}\mathbf{x}, (1-\bar{\alpha}_t)\mathbf{I}\right], \tag{2.28}$$

allowing for arbitrary noisy latents to be sampled directly from $\mathbf{x}$ without simulating all previous steps.

The reverse transitions $q(\mathbf{z}_{t-1}|\mathbf{z}_t)$, however, cannot be determined analytically due to the unknown data distribution. Instead, DDPM models learn a Gaussian approximation

$$p_\phi(\mathbf{z}_{t-1} \mid \mathbf{z}_t) = \text{Norm}_{\mathbf{z}_{t-1}}\left[\mathbf{f}_\phi(\mathbf{z}_t, t), \sigma_t^2\mathbf{I}\right]. \tag{2.29}$$

Chaining many of these denoising steps allows for construction of the joint distribution, $p_\phi(\mathbf{x}, \mathbf{z}_{1,\ldots,T})$, which can then be used to derive the evidence lower bound, ELBO such that

$$\log p_\phi(\mathbf{x}) \geq \text{ELBO}(\mathbf{x}; \phi). \tag{2.30}$$

Minimising the negative ELBO therefore trains a network to predict slightly denoised latents $\mathbf{z}_{t-1}$ from the noisier $\mathbf{z}_t$ versions.

Using Bayes theorem along with the Gaussian change of variables identity and the product of two Gaussians combination identity allows us to express

$$q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = \frac{q(\mathbf{z}_t|\mathbf{z}_{t-1}, \mathbf{x})q(\mathbf{z}_{t-1}|\mathbf{x})}{q(\mathbf{z}_t|\mathbf{x})} \tag{2.31}$$

$$= \text{Norm}_{\mathbf{z}_t}\left[\sqrt{1-\beta_t}\mathbf{z}_{t-1}, \beta_t\mathbf{I}\right] \text{Norm}_{\mathbf{z}_{t-1}}\left[\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}, (1-\bar{\alpha}_{t-1})\mathbf{I}\right] \tag{2.32}$$

$$= \text{Norm}_{\mathbf{z}_{t-1}}\left[\frac{(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\sqrt{1-\beta_t}\mathbf{z}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}, \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{I}\right]. \tag{2.33}$$

Therefore, if the clean image $\mathbf{x}$ were known, the exact reverse transition would also be known. We can therefore equivalently train a denoising model to predict clean images, conditioned on the noisy image and the noise level, $\hat{\mathbf{x}}_\phi$, such that

$$\mathbf{x} \approx \hat{\mathbf{x}}_\phi(\mathbf{z}_t, t). \tag{2.34}$$

When used in conjunction with (2.33), this provides a local denoising direction, allowing repeated movement to slightly cleaner states.

Equation (2.28) also shows that predicting **x** is equivalent to predicting the noise added to it at a given diffusion step, so

$$\varepsilon_\phi(\mathbf{z}_t, t) = \frac{\mathbf{z}_t - \sqrt{\bar{\alpha}_t}\hat{\mathbf{x}}_\phi(\mathbf{z}_t, t)}{\sqrt{1-\bar{\alpha}_t}}. \tag{2.35}$$

In this formulation, we can simplify the ELBO with Gaussian identities and the properties of the KL divergence, giving rise to the standard DDPM training loss [],

$$\mathscr{L}_{\text{DDPM}}(\phi) = \mathbb{E}_{\mathbf{x},t,\varepsilon}\left[\left\|\varepsilon - \varepsilon_\phi\left(\sqrt{\bar{\alpha}_t}\mathbf{x} + \sqrt{1-\bar{\alpha}_t}\varepsilon, t\right)\right\|_2^2\right]. \tag{2.36}$$

Applying Tweedie's formula [] to (2.28), and applying the known form of a Gaussian mean, we can then also show that

$$\nabla_{\mathbf{z}_t} \log p_t(\mathbf{z}_t) \approx -\frac{\varepsilon_\phi(\mathbf{z}_t, t)}{\sqrt{1-\bar{\alpha}_t}}, \tag{2.37}$$

where we notice the left hand term is equal to the score of the distribution of noisy latents at time t. In general, we define this as

$$\mathbf{s}_t(\mathbf{z}) = \nabla_{\mathbf{z}} \log p_t(\mathbf{z}). \tag{2.38}$$

Therefore, predicting the slightly denoised latent, $\mathbf{z}_{t-1}$, the fully denoised image, **x**, the noise, ε, and the score, $\nabla_{\mathbf{z}} \log p_t(\mathbf{z})$ are equivalent parameterisations of the same reverse process. For inverse problems, the score parameterisation is particularly useful because it gives a learned prior gradient, which can be exploited in conjunction with measurement-consistency terms from forward models.

2.2.1 Score-Based SDEs and Variance-Exploding Models

In the continuous-time limit, we can write a noising process as the stochastic differential equation,

$$d\mathbf{z} = \mathbf{f}(\mathbf{z}, \tau)d\tau + g(\tau)d\mathbf{w}, \tag{2.39}$$

where $\tau \in [0,1]$ denotes the continuous diffusion time, and $\mathbf{w}$ is a Wiener process []. Anderson and Song [] have shown that where $p_\tau(\mathbf{z})$ denotes the marginal distribution of the noisy latents,

$$p_\tau(\mathbf{z}) = \int p_\tau(\mathbf{z}|\mathbf{x}) p_{\text{data}}(\mathbf{x})\, d\mathbf{x}, \tag{2.40}$$

the reverse-time process is

$$d\mathbf{z} = \left[\mathbf{f}(\mathbf{z},\tau) - g(\tau)^2 \nabla_{\mathbf{z}} \log p_\tau(\mathbf{z})\right] d\tau + g(\tau) d\bar{\mathbf{w}}, \tag{2.41}$$

where time is integrated backwards and $d\bar{\mathbf{w}}$ is the reverse-time Wiener process. Therefore, once the score, $\nabla_{\mathbf{z}} \log p_\tau(\mathbf{z})$ is known, the noising process can be approximately reversed, facilitating sampling of the learned image distribution.

The choices of $\mathbf{f}$ and g in (2.39) define different families of SDEs. Variance-preserving models are the continuous-time analogue of the DDPM construction above. They may be written as

$$d\mathbf{z} = -\frac{1}{2}\beta(\tau)\mathbf{z}\, d\tau + \sqrt{\beta(\tau)}\, d\mathbf{w}, \tag{2.42}$$

where $\beta(\tau)$ is the continuous-time noise schedule - the continuous analogue of the discrete DDPM schedule β_t.

In contrast, variance-exploding (VE) SDEs are parameterised directly by a noise scale σ, and add Gaussian noise with increasing variance, such that a noisy image at noise level σ is given by

$$\mathbf{z}_\sigma = \mathbf{x} + \sigma\varepsilon, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{2.43}$$

so the marginal probability to use in the reverse-time process becomes

$$p_\sigma(\mathbf{z}) = \int \mathcal{N}(\mathbf{z}; \mathbf{x}, \sigma^2\mathbf{I}) p_{\text{data}}(\mathbf{x})\, d\mathbf{x}. \tag{2.44}$$

We can then train a denoiser conditioned on this noise level, $D_\phi(\mathbf{z},\sigma)$ to recover clean images from their noisy versions using the loss,

$$\mathscr{L}_{\mathrm{VE}}(\phi) = \mathbb{E}_{\mathbf{x},\sigma,\varepsilon}\left[\lambda(\sigma)\left\|D_\phi(\mathbf{x}+\sigma\varepsilon,\sigma)-\mathbf{x}\right\|_2^2\right], \tag{2.45}$$

where $\lambda(\sigma)$ is a weighting over noise scales which can be used to ensure losses at all scales contribute similarly. As the best predictor of a random variable under squared-error loss is its conditional expectation, this also implies that the optimal denoiser is such that

$$D^*(\mathbf{z},\sigma) = \mathbb{E}[\mathbf{x} \mid \mathbf{z}]. \tag{2.46}$$

Therefore, using Tweedie's formula gives

$$\nabla_{\mathbf{z}} \log p_\sigma(\mathbf{z}) = \frac{\mathbb{E}[\mathbf{x} \mid \mathbf{z}] - \mathbf{z}}{\sigma^2} \tag{2.47}$$

$$\implies \mathbf{s}_\phi(\mathbf{z},\sigma) = \frac{D_\phi(\mathbf{z},\sigma)-\mathbf{z}}{\sigma^2}, \tag{2.48}$$

allowing denoising models to be used in both reverse-VE-SDE samplers and inverse-problem methods as a prior score.

2.2.2 The Suitability of Variance-Exploding Models for CT Imaging

The VE diffusion approach is a natural fit for medical imaging inverse problems because the addition of Gaussian noise to form latent images preserves the units of the underlying images. This is in contrast with VP noising processes, which work by both attenuating the image and adding noise, as demonstrated for DDPM in (2.24).

We can therefore interpret a VP-denoised prediction, $D_\phi(\mathbf{z},\sigma)$ as a clean-image estimate, enabling direct application of medical imaging measurement models, which are defined on the clean images. For example, in CT reconstruction, we can predict the corresponding sinogram as $AD_\phi(\mathbf{z},\sigma)$, where A is the CT forward operator [**? ? ?**].

2.2.3 EDM Preconditioning

In some cases, we choose not to use a denoising model directly, but instead embed one within a predconditioning framework. This is the case for EDM [**?**], which wraps a raw neural network, F_ϕ, with scale-dependent input, output and skip coefficients to form a preconditioned denoiser. This gives the raw network a better conditioned learning problem because inputs are normalised to have a similar scale across noise levels, and the network only needs to predict the part of the denoising map not already captured by the skip connection. For a noisy image $\mathbf{z}$ at noise level σ, this gives

$$D_\phi(\mathbf{z},\sigma) = c_{\text{skip}}(\sigma)\mathbf{z} + c_{\text{out}}(\sigma)F_\phi\left(c_{\text{in}}(\sigma)\mathbf{z}, c_{\text{noise}}(\sigma)\right), \tag{2.49}$$

where

$$c_{\text{in}}(\sigma) = \frac{1}{\sqrt{\sigma^2+\sigma_{\text{data}}^2}}, \qquad c_{\text{skip}}(\sigma) = \frac{\sigma_{\text{data}}^2}{\sigma^2+\sigma_{\text{data}}^2}, \tag{2.50}$$

$$c_{\text{out}}(\sigma) = \frac{\sigma\sigma_{\text{data}}}{\sqrt{\sigma^2+\sigma_{\text{data}}^2}}, \qquad c_{\text{noise}}(\sigma) = \frac{\log\sigma}{4}, \tag{2.51}$$

and σ_{data} is an estimate of the standard deviation of clean training images. The coefficient c_{in} rescales the noisy input before it is passed to F_ϕ, while c_{noise} convets the noise level to a logarithmic scale.

The skip connection allows the denoiser make only small adjustments at low noise levels, whereas at high noise levels, c_{skip} becomes small, so the denoised estimate relies more heavily on the learned network output. The coefficient c_{out} then scales this output so that it has the appropriate magnitude for the current noise level. Once the wrapped denoiser D_ϕ has been formed, a score estimate can still be given by Equation 2.48.

2.2.4 The PaDIS Approach

Instead of learning to denoise complete images, PaDIS learns a diffusion prior from image patches, alongside their positional information. This aims to reduce both the memory and data requirements for training, with single images providing many training examples [**?**].

Let G_c be an operator which extracts a patch, c, of size P from an image $\mathbf{x}$ of size N, along with the positional encoding for the patch, ρ_c, before adding noise to only the image part. This positional encoding is formed by scaling the x and y coordinates of the image pixels and scaling them between -1 and 1. A noisy patch can then be denoted as

$$G_c\mathbf{x} + \sigma\varepsilon, \qquad \varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{2.52}$$

Using this, Hu et. al. define a patch denoising objective,

$$\mathcal{L}_{\text{PaDIS}}(\phi) = \mathbb{E}_{\mathbf{x},c,\sigma,\varepsilon}\left[\left\|D_\phi(G_c\mathbf{x} + \sigma\varepsilon, \rho_c, \sigma) - G_c\mathbf{x}\right\|_2^2\right], \tag{2.53}$$

which according to (2.48) gives the patch score,

$$\mathbf{s}_{\phi,c}(G_c\mathbf{z}, \rho_c, \sigma) = \frac{D_\phi(G_c\mathbf{z}, \rho_c, \sigma) - G_c\mathbf{z}}{\sigma^2}. \tag{2.54}$$

Where $k = \lfloor N/P \rfloor$, an $N \times N$ image could be covered by a $(k+1) \times (k+1)$ grid of non-overlapping $P \times P$ patches after padding. Naively fixing this patch partition and combining the corresponding patch scores would enable construction of a whole-image score, but would cause visible patch boundaries because the patch boundaries remain fixed throughout sampling. To mitigate this, PaDIS zero-pads each image on all sides by

$$M = (k+1)P - N, \tag{2.55}$$

giving a padded image of size $(N+2M) \times (N+2M)$. It then considers many different non-overlapping patch partitions, indexed by shifts

$$(i, j) \in \{1, \ldots, M\}^2. \tag{2.56}$$

Each choice of (i, j) also implicitly defines the remaining outer border region from zero-padding. This adds flexibility to the model, allowing us to sample from different shifts during reconstruction to reduce the fixed boundary artefacts.

Using the above, the full-image distribution at noise level σ can be modelled as

$$p_\sigma(\mathbf{z}) = \frac{1}{Z_\sigma} \prod_{i,j} p_{\sigma,i,j,B}(\mathbf{z}_{i,j,B}) \prod_{r=1}^{(k+1)^2} p_{\sigma,i,j,r}(\mathbf{z}_{i,j,r}), \tag{2.57}$$

where $\mathbf{z}_{i,j,r}$ denotes the rth patch under shift (i, j), $\mathbf{z}_{i,j,B}$ denotes the bordering zero-padded region, and Z_σ is a normalising constant. Differentiating the logarithm of this distribution then gives the whole-image score as a sum of patch scores;

$$\nabla_{\mathbf{z}} \log p_\sigma(\mathbf{z}) = \sum_{i,j} \left(\mathbf{s}_{\sigma,i,j,B}(\mathbf{z}_{i,j,B}) + \sum_{r=1}^{(k+1)^2} \mathbf{s}_{\sigma,i,j,r}(\mathbf{z}_{i,j,r}) \right). \tag{2.58}$$

The border term is zero by construction, and the patch terms are learned by the denoising network. Computing this over all possible partitions at each sampling step would be prohibitively expensive, so sampling and reconstruction instead select one random shift at each iteration, and use this to estimate the whole-image score. For a selected shift (i, j), this gives the PaDIS score estimate

$$\mathbf{s}_{\text{PaDIS}}(\mathbf{z}, \sigma) \approx \sum_{r=1}^{(k+1)^2} G_{i,j,r}^T \mathbf{s}_{\phi,i,j,r}(G_{i,j,r}\mathbf{z}, \rho_{i,j,r}, \sigma), \tag{2.59}$$

where $G_{i,j,r}$ extracts the rth patch under the selected partition and $G_{i,j,r}^T$ places the corresponding patch score back into the full image. The boundary artefacts are then instead removed through the iterations, as different random shifts are used at different sampling steps. The flexibility of this prior and its ability to approximate the whole-image score allow it to be used with any standard diffusion sampling and inverse-problem method. The PaDIS paper found that combining diffusion posterior sampling (DPS) and Langevin dynamics worked best for their CT reconstructions.

2.3 Reconstruction Methods

2.3.1 Filtered Back-Projection and the FDK Method

Within the parallel-beam regime, filtered back-projection (FBP) can be used as a direct reconstruction method. It works by converting the measured sinogram into radial slices of the Fourier domain representation of $f(x,y)$ using the Fourier slice theorem. To convert this into the image domain, we perform an inverse Fourier transform. However, to prevent a need for interpolation, we perform this transform in polar coordinates, introducing a Jacobian factor $|\omega|$. This factor acts as a ramp [**?**] filter, and corrects the over-representation of low-frequencies which would otherwise dominate the signal. This gives frequency-weighted projections,

$$Q_\theta(t) = \int_{-\infty}^{\infty} S_\theta(\omega)|\omega|e^{2\pi i\omega t}\,d\omega. \tag{2.60}$$

This gives the desired image, but in the detector coordinate system (θ, t). To recover the (x, y)-space image, we back-project into the image domain using

$$\hat{f}(x,y) = \int_0^{\pi} Q_{\theta}(x\cos\theta + y\sin\theta)\, d\theta, \tag{2.61}$$

or with a finite number of views,

$$\hat{f}(x,y) \approx \frac{\pi}{N_{\theta}} \sum_{i=1}^{N_{\theta}} Q_{\theta_i}(x\cos\theta_i + y\sin\theta_i). \tag{2.62}$$

This method is fast and interpretable, but requires dense angular sampling to achieve good reconstructions.

These same ideas can be extended to realistic fan-beam and cone-beam geometries. These methods still filter projection data and back-project into the image domain, but must also account for the divergent ray geometries. In circular cone-beam CT, this leads to the Feldkamp-Davis-Kress (FDK) algorithm, which is typically used in industry to approximate circular cone-beam reconstructions [].

2.3.2 Iterative Reconstruction and Denoising Priors

Instead of trying to perform reconstruction in one analytical operation, iterative reconstructions treat reconstruction as a data-fitting problem, repeatedly updating an initial image estimate to make its forward projection better match the measured sinogram. For a forward model

$$\mathbf{y} = A\mathbf{x} + \eta, \tag{2.63}$$

this often involves minimising the squared residual,

$$\mathscr{L}_{\mathbf{y}}(\mathbf{x}) = \frac{1}{2}\|\mathbf{y} - A\mathbf{x}\|_2^2, \tag{2.64}$$

The gradient of this loss is

$$\nabla_{\mathbf{x}}\mathscr{L}_{\mathbf{y}}(\mathbf{x}) = A^T(A\mathbf{x} - \mathbf{y}), \tag{2.65}$$

enabling data-consistency updates of the form

$$\mathbf{x}^{k+1} = \mathbf{x}^k - \delta_{\text{data}} A^T (A\mathbf{x}^k - \mathbf{y}), \tag{2.66}$$

which locally reduce the projection mismatch. The scaling of the update step, δ_{data}, however, cannot be chosen independently of the forward operator because the scale of A depends on the reconstruction geometry and image normalisation. To account for this, we can use the operator Hessian to find the residual curvature,

$$\kappa = \nabla^2_{\mathbf{x}} \mathscr{L}_{\mathbf{y}}(\mathbf{x}) = A^T A, \tag{2.67}$$

meaning that for the pure quadratic loss term, we can produce stable updates in all directions by taking

$$0 < \delta_{\text{data}} < \frac{2}{\|A^T A\|_2}, \tag{2.68}$$

where

$$\|A^T A\|_2 = \max_{\Delta\mathbf{x} \neq 0} \frac{\|A^T A \Delta\mathbf{x}\|_2}{\|\Delta\mathbf{x}\|_2}. \tag{2.69}$$

Observing that $A^T A$ is symmetric and positive semi-definite then allows us to set

$$\|A^T A\|_2 = \lambda_{\max}(A^T A) = \|A\|_2^2, \tag{2.70}$$

so a more stable version of the update is

$$\mathbf{x}^{k+1} = \mathbf{x}^k - \frac{\zeta}{\|A\|_2^2} A^T (A\mathbf{x}^k - \mathbf{y}), \tag{2.71}$$

with $0 < \zeta < 2$ giving stability. Although this does not guarantee convergence once additional priors, denoisers or sampling steps are introduced, it does give the measurement-gradient part of the update the correct scale. This scaling is particularly important in CT because $\|A\|_2$ depends on the projection geometry, detector spacing, number of views, image resolution and measurement units. Therefore, without it, a specific data-consistency coefficient loses meaning across different setups.

This squared error term alone is normally insufficient In sparse-view CT, because many images provide approximately correct measurement projections. Iterative reconstruction is therefore often combined with other regularising terms, $R(\mathbf{x})$ such that

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \frac{1}{2}\|\mathbf{y} - A\mathbf{x}\|_2^2 + \lambda R(\mathbf{x}). \tag{2.72}$$

Often, a total-variation regulariser is used, where

$$R(\mathbf{x}) = \mathrm{TV}(\mathbf{x}) = \sum_i \sqrt{(D_x\mathbf{x})_i^2 + (D_y\mathbf{x})_i^2 + \varepsilon^2}, \tag{2.73}$$

favouring piecewise smooth images whilst still allowing share edges, approximately matching the expected form for CT images.

Instead of these explicit regularisers, Plug-and-play methods utilise a learned denoising step and alternate between data-consistence and denoising updates, often with

$$\tilde{\mathbf{x}}^{k+1} = \mathbf{x}^k - \delta_{\text{data}} A^T (A\mathbf{x}^k - \mathbf{y}), \tag{2.74}$$

followed by

$$\mathbf{x}^{k+1} = D_\sigma(\tilde{\mathbf{x}}^{k+1}), \tag{2.75}$$

where D_σ has been trained to remove noise at scale σ. This allows the data-consistency step to keep reconstructions near the measured sinogram, whilst the denoising step directs updates towards the distribution of clean training images.

2.3.3 Diffusion Models for Inverse Problems

As discussed in Section 2.2.1, we can use a variance exploding denoiser to estimate the score. At a fixed noise level, σ_k, this can be used to inform a Langevin update to an existing reconstruction $\tilde{\mathbf{z}}^k$ as

$$\mathbf{z}^{k+1} = \tilde{\mathbf{z}}^k + \frac{\alpha_k}{2}\mathbf{s}_\phi(\tilde{\mathbf{z}}^k, \sigma_k) + \sqrt{\alpha_k}\xi^k, \qquad \xi^k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{2.76}$$

Here, the score term moves the reconstruction towards regions of higher probability under the learned image prior, while the stochastic term maintains sampling diversity.

To move effectively perform reconstruction, we can combine this prior update with a data consistency update by inserting a gradient step against the measurement residual,

$$\tilde{\mathbf{z}}^{k+1} = \mathbf{z}^k + \delta_{\text{data}} A^T(\mathbf{y} - A\mathbf{z}^k), \tag{2.77}$$

followed by at least one Langevin prior step from (2.76). Here, the data step guides the image towards consistency with the measured sinogram, and the Langevin step pulls it towards the learned image prior. We will refer to this as the Langevin reconstruction method throughout.

Diffusion posterior sampling (DPS) instead applies the sinogram measurement model to the denoised estimate, such that if we define the clean image as

$$\hat{\mathbf{x}}_0^k = D_\phi(\mathbf{z}^k, \sigma_k), \tag{2.78}$$

then we can construct a data consistency update with

$$\tilde{\mathbf{z}}^{k+1} = \mathbf{z}^k - \frac{\zeta_k}{2}\nabla_{\mathbf{z}^k}\|\mathbf{y} - A\hat{\mathbf{x}}_0^k\|_2^2, \tag{2.79}$$

which can then again be followed by at least one Langevin prior step as in (2.76). This will generally be referred to as DPS-Langevin reconstruction. However, when we use the PaDIS score computation Equation 2.59, it will be referred to as the PaDIS reconstruction method.

By discretising the reverse-time SDE (2.41), we can derive an alternative prediction step moving from the defined noise level σ_k to σ_{k+1} as

$$\tilde{\mathbf{z}}^{k+1} = \mathbf{z}^k + (\sigma_{k+1}^2 - \sigma_k^2)\mathbf{s}_\phi(\mathbf{z}^k, \sigma_k) + \sqrt{\sigma_{k+1}^2 - \sigma_k^2}\,\xi^k, \tag{2.80}$$

which we can then again correct towards the regions of higher probability under the learned image prior using the same Langevin step from (2.76). This is known as the predictor-corrector (PC) method.

Instead of descending the square residual, DDNM methods impose measurement consistency through an approximate pseudoinverse of the form

$$\hat{\mathbf{x}}_{\mathbf{y}}^k = A^\dagger\mathbf{y} + (I - A^\dagger A)\hat{\mathbf{x}}_0^k. \tag{2.81}$$

Here, the first term gives the component determined by the sinogram, and the second acts to preserve the component of the denoised estimate lying approximately within the null space of the forward model. Within the VE implementation, this can also be converted back to a score estimate via

$$\mathbf{s}_{\text{DDNM}}(\mathbf{z}^k, \sigma_k) = \frac{\hat{\mathbf{x}}_{\mathbf{y}}^k - \mathbf{z}^k}{\sigma_k^2}. \tag{2.82}$$

The reconstruction is then, again updated at least once using the Langevin step from (2.76), where $\mathbf{s}_\phi(\tilde{\mathbf{z}}^k, \sigma_k) = \mathbf{s}_{\text{DDNM}}(\mathbf{z}^k, \sigma_k)$.

3 Methods

3.1 The Dataset

In this work, we use the LIDC-IDRI dataset [?] for all experiments. This dataset contains 1018 human chest CT scans as raw CT DICOM [?] images from 1010 patients, with each scan containing many axial slices, and some slices containing cancerous nodules. Before training or reconstruction, we convert these into a processed slice dataset, saving each slice as a 512×512 NumPy array, along with a nodule mask. All but the 512-resolution experiments then downsample these to 256×256 resoution during dataloading. Although the dataset contains multiple scans for some patients, the LION processing pipeline keeps only one per pateient, preventing accidental data leakage.

Divisioning into train, validation and test sets is done by sorted patient ID in an 80:10:10 ratio, ensuring that slices from no patient are present across sets. The default PaDIS training regime then selects at most 4 slices per patient, targeting a balance between nodule-containing and nodule-free slices. This is, however, constrained by the availability of annotations for each patient, so some contribute fewer than 4 slices. The detailed breakdown is presented in **??**.

Split	Patient IDs	Images in 4-slice regime	Images in all-slice regime
Train	809	2713	189725
Validation	101	328	27426
Test	102	326	25719
Total	1012	3367	242870

Table 3.1 Processed LIDC-IDRI split sizes used in this work. Note that the dataset contains two unused patient IDs.

3.2 Experimental Geometry

Throughout all experiment, we use the fan-beam approach explained in Equation 2.3

3.3 Model Training

In this work, we train three classes of model to aid reconstruction; patch-based diffusion models, whole-image diffusion models, and PnP denoisers.

3.3.1 Diffusion Model Training

The learned diffusion priors were trained following the variance-exploding regime described in subsection 2.2.1, and their denoisers can therefore be converted to scores and used for both unconditional generation and solving inverse problems as explained in **??**.

All of the diffusion denoisers were trained using the U-Net like, noise-conditional NSCN++ model, F_ϕ, developed by Song et. al. [**?**], which allows multi-scale inputs, and predicts denoised images directly. This network passes the noise level to each residual block, and always takes three channel inputs - the noisy image, and the [-1,1] normalised x- and y-coordinates of each pixel. Throughout, we use dropout level 0.05, a base attention channel size of 128, 4 residual blocks per level, and attention resolution 16. These models always use four attention levels, with the patch-based models using 128, 256, 256, and 256 channels at each level, and the whole-image models using 128, 256, 256, and 512 channels for greater coarse-level capacity.

All diffusion models were trained using the Adam optimiser [**?**], with a learning rate of 2×10^{-4} for diffusion models, and the EDM preconditioning described in subsection 2.2.3 with $\sigma_{\text{data}} = 0.5$. The whole-image models were trained for 24 hours, and patch-based models trained for 12 hours on an RTX6000 Pro GPU. In both cases, the learning-rate ramped up over the first 10 million seen training examples or patches. During diffusion model training, noise scales were sampled from an EDM-style log-normal distribution, and truncated to $\sigma \in [0.002, 40]$, with $\log\sigma \sim \mathcal{N}(-1.2, 1.2^2)$.

Using these conventions, whole-image diffuson denoisers were trained to denoise entire 256×256 CT slices, and patch-based denoisers were trained for images of size 256×256 and 512×512 with patch sizes 56×56, 32×32, and 16×16, where selection probabilities for these at each training step are 0.5, 0.3, and 0.2 respectively. Ablation models were also trained using all available processed slices, different patch sizes, and without positional encoding channels. This patch size schedule is used in all experiments apart from those for the patch size ablations, which are detailed in Table 3.2.

The diffusion denoisers were evaluated on the validation set throughout training to ensure overfitting did not occur. However, for the final six hours of each run, the validation frequency and intensity was significantly increased, and the checkpoint correspoinding to the minimum validaton loss in this period was used as the final model.

Ablation	Sampled patch sizes	Sampling probabilities	Padding width
$P = 8$	8	1.0	8
$P = 16$	8, 16	0.3, 0.7	16
$P = 32$	8, 16, 32	0.2, 0.3, 0.5	32
$P = 56$	16, 32,6	0.2, 0.3, 0.5	24
$P = 96$	32, 64, 96	0.2, 0.3, 0.5	32

Table 3.2 Patch-size schedules used for the PaDIS patch-size ablations. Each training step samples one patch size according to the listed probabilities. The $P = 56$ row is the default PaDIS patch schedule used for the main 256-resolution patch model.

3.3.2 PnP Denoiser Training

Two PnP denoisers were trained using the existing PnP-ADMM [1] LION implementation, which uses a DRUNet model [5] as the denoiser. At each training step, Gaussian noise is sampled uniformly from $\sigma \in [0, 0.05]$ and added to the clean image, from which the model then learns to recover the clean target.

Each model was trained using the Adam optimiser, a learning rate of 1×10^{-4} for 100 epochs, batch size 8, and MSE denoising loss. One model was trained with only the noisy image as input, and another was trained with the noise level additionally provided as a constant image channel. Both used 64 internal chanels, 4 blocks per level, bias-free convolutions, and leaky-ReLU activations. Validation was performed after each epoch using the same Gaussian denoising objective, and for each model, the checkpoint corresponding to the lowest validation loss was used for inference experiments.

3.4 Reconstructor Implementation

- Implemented in LION - Produced three implementation tracks: Explicit paper implementation, public-repo matching setup, and one which uses correct operator normalisations to increase cross-problem reliability, as described in **??**. - Created LION-physics implementations for all experiments, but only paper and public-repo matching implementations for those which were actually described in sufficient detail for reproduction in those respective soruces. This is summarised in **??**.

Table 3.3 Reconstruction methods which were possible to implement in the same manner as described in the original PaDIS paper, or their publicly available codebase, as well as those which were implemented in our customised LION-physics framework.

Method	Paper-style	Public-compatible	LION-physics
Baseline FDK	–	–	✓
ADMM-TV	–	–	✓
PnP-ADMM	–	–	✓
Whole-image diffusion	✓	–	✓
Langevin	✓	✓	✓
Predictor-corrector	✓	✓	✓
VE-DDNM	✓	✓	✓
Patch averaging	–	✓	✓
Patch stitching	–	✓	✓
PaDIS-DPS	✓	✓	✓

3.4.1 Public-Compatible Implementations

- include correctness checks

3.4.2 LION-Physics Implementations

3.4.3 Hyperparameter Tuning

3.5 Experiments

a description of the different experiments and their purpose, flagging any differences from the original padis paper.

3.6 Metrics

3.7 Compute and Reproducibility

4 Results

5 Discussion

6 Conclusion

References

[1] Chan, S. H., Wang, X., and Elgendy, O. A. (2016). Plug-and-Play ADMM for Image Restoration: Fixed Point Convergence and Applications.

[2] Ho, J., Jain, A., and Abbeel, P. (2020). Denoising Diffusion Probabilistic Models.

[3] Özdenizci, O. and Legenstein, R. (2023). Restoring Vision in Adverse Weather Conditions With Patch-Based Denoising Diffusion Models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(8):10346–10357.

[4] Rumberger, J. L., Yu, X., Hirsch, P., Dohmen, M., Guarino, V. E., Mokarian, A., Mais, L., Funke, J., and Kainmueller, D. (2021). How Shift Equivariance Impacts Metric Learning for Instance Segmentation. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7108–7116, Montreal, QC, Canada. IEEE.

[5] Zhang, K., Li, Y., Zuo, W., Zhang, L., Gool, L. V., and Timofte, R. (2021). Plug-and-Play Image Restoration with Deep Denoiser Prior.

A LLM Usage Declaration

LLMs were used for assistance with coding, as declared in the repository README file. They were never used to generate text for inclusion in this document, and were only used to proofread and suggest minor wording alterations.

B Additional Figures